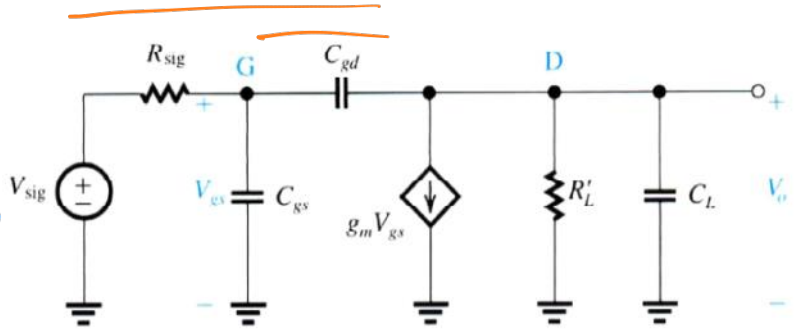
Reading

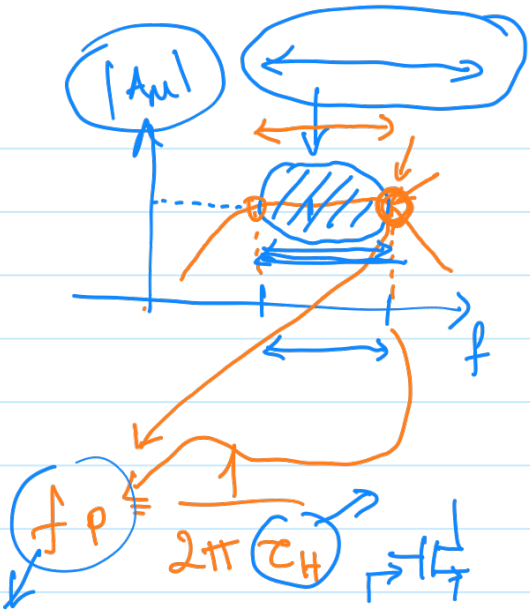
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.4 High-Frequency Response of Common-Gate and Cascode Amplifiers
- Chapter 10.5 High-Frequency Response of Source and Emitter Fallowers
- Chapter 10.6 High-Frequency Response of Differential Amplifiers

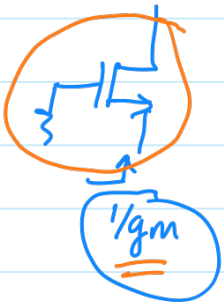
Recap

Common Source Amplifier

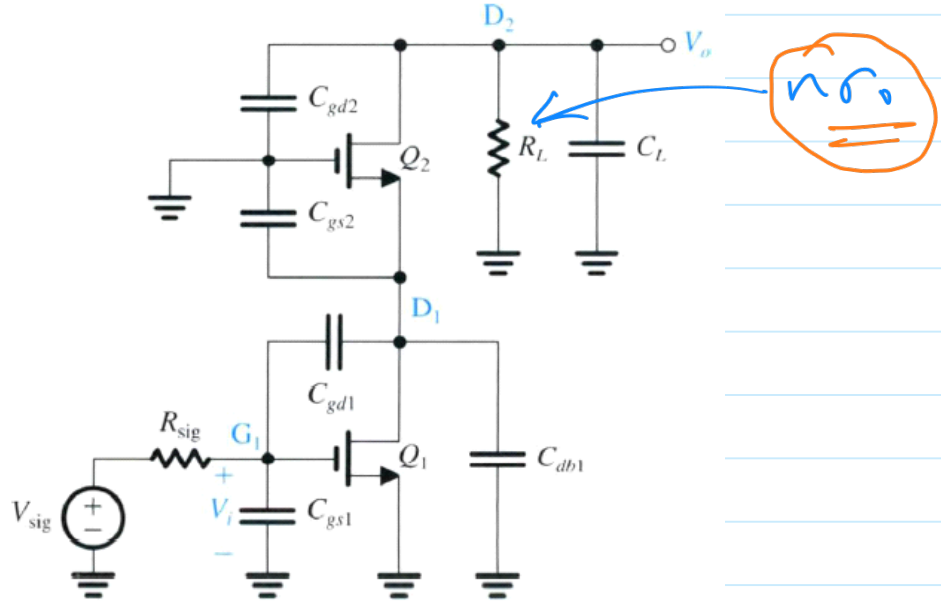




$$\tau_H = C_{gs}R_{gs} + C_{gd}R_{gd} + C_L R_{C_L}$$
$$= C_{gs}R_{\text{sig}} + C_{gd}[R_{\text{sig}}(1 + g_m R'_L) + R'_L] + C_L R'_L$$

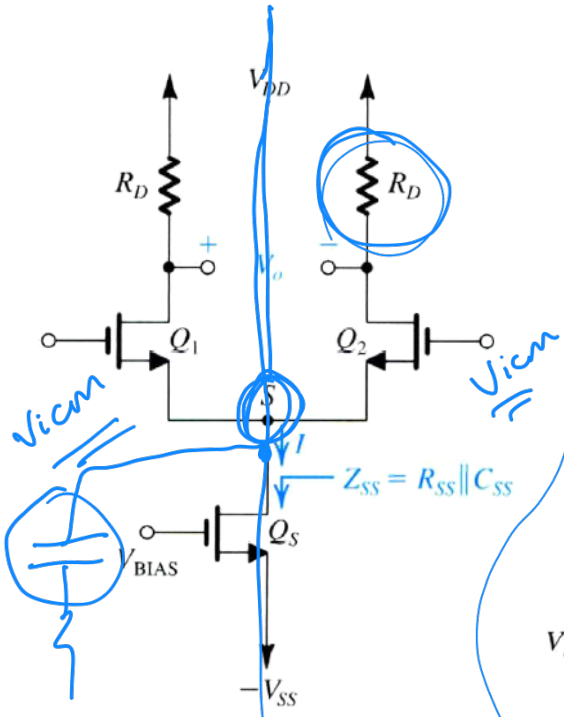


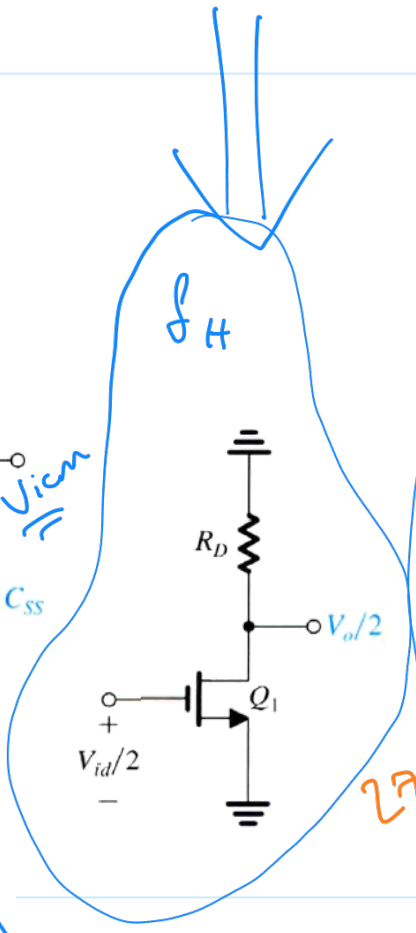
Cascode Amplifier

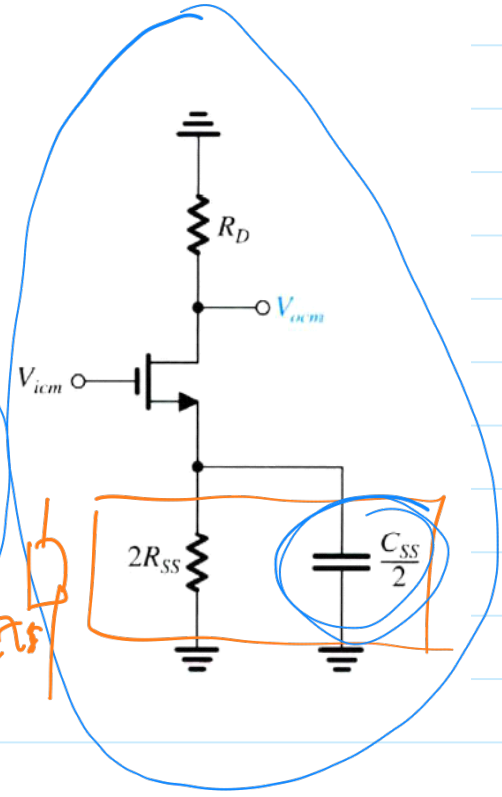


$$\tau_H = C_{gs1}R_{\text{sig}} + C_{gd1}[(1 + g_{m1}R_{d1})R_{\text{sig}} + R_{d1}]$$
$$+ (C_{db1} + C_{gs2})R_{d1} + (C_L + C_{gd2})(R_L \| R_o)$$

Recap



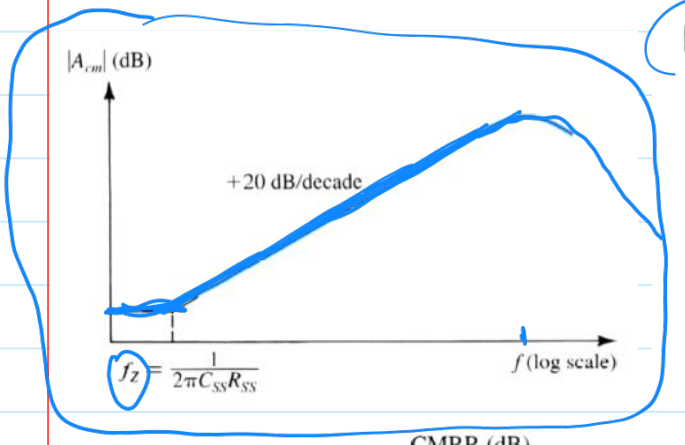


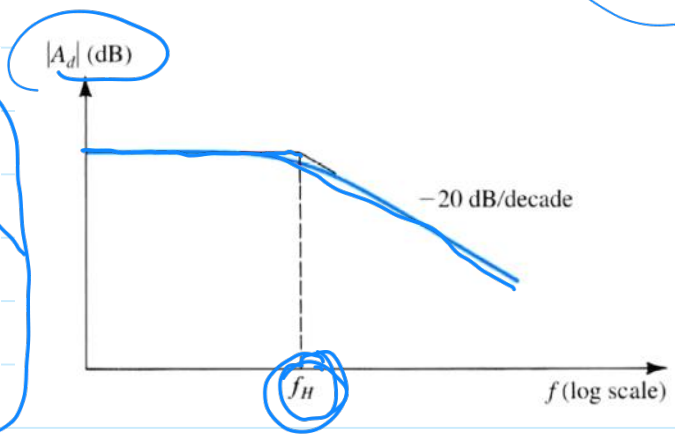


$$A_{cm} = -\left(\frac{R_D}{2R_{SS}}\right)\frac{\Delta R_D}{R_D}$$

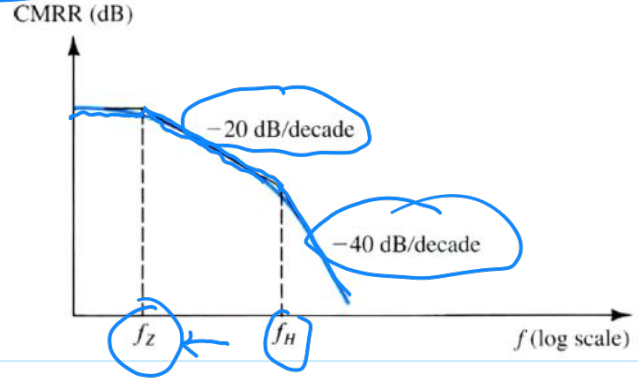
$$A_{cm}(s) = -\frac{R_D}{2R_{SS}}\left(\frac{\Delta R_D}{R_D}\right)(1 + sC_{SS}R_{SS})$$

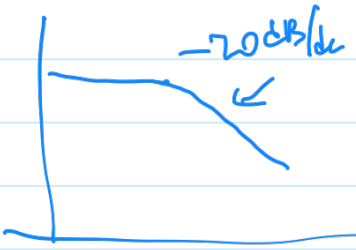
$$f_z = \frac{\omega_Z}{2\pi} = \frac{1}{2\pi C_{SS}R_{SS}}$$





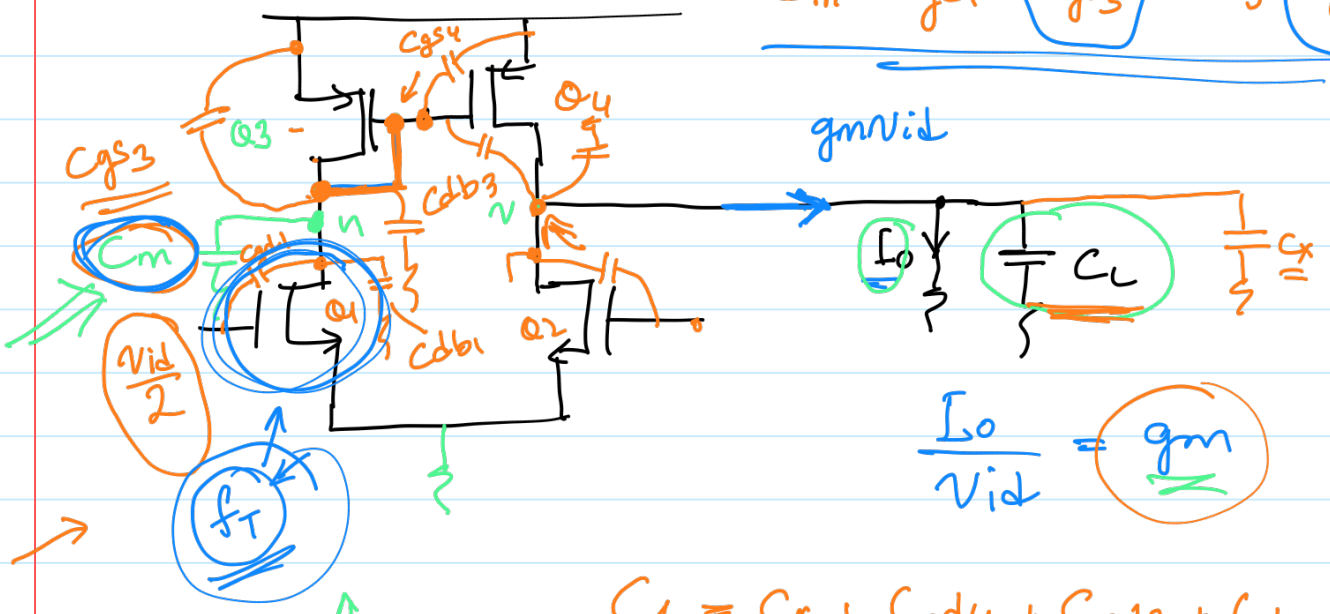
$\frac{|A_d|}{|A_{cm}|}$





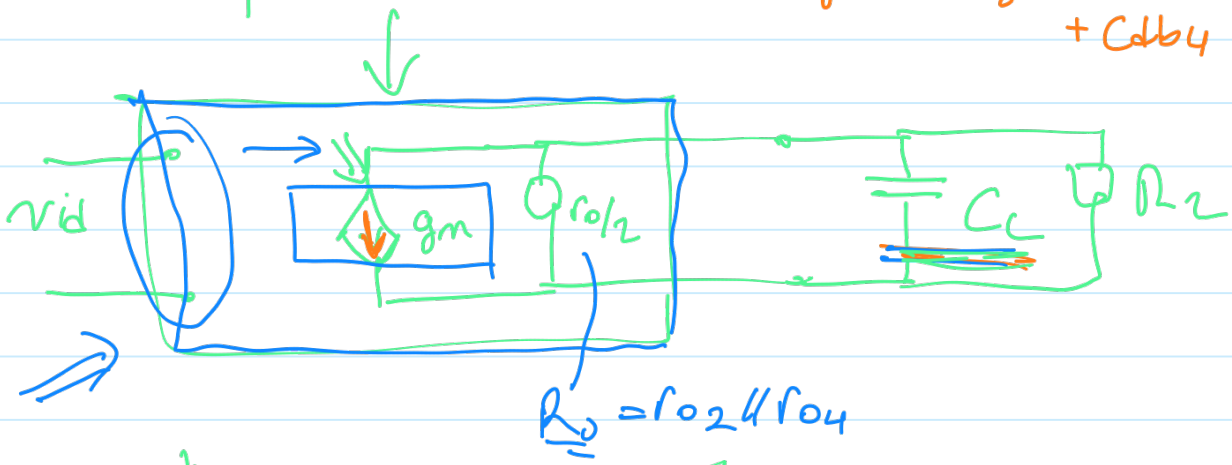
HF Resp. of CM Loaded Differential Amp.

$$C_m = C_{gd1} + C_{gs3} + C_{db3} + C_{gs4} + C_{db4}$$



$$\frac{I_o}{V_{id}} = g_m$$

$$C_L = C_x + C_{gd4} + C_{gd2} + C_{db2} + C_{db4}$$



$$R_o = r_{o2} \| r_{o4}$$

$$f_{p1} = \frac{1}{2\pi . C_L R_o}$$

$$G_m = \frac{I_o}{V_{id}} = g_m \frac{1 + s\frac{C_m}{2g_{m3}}}{1 + s\frac{C_m}{g_{m3}}}$$

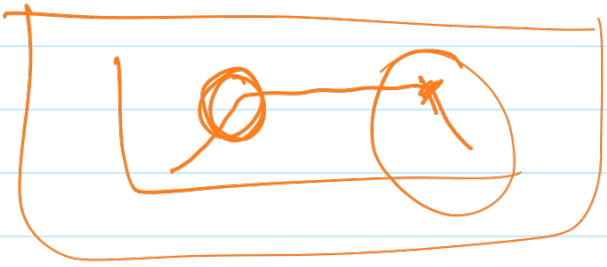
$$f_{p2} = \frac{g_{m3}}{2\pi C_m} \qquad f_z = \frac{2g_{m3}}{2\pi C_m}$$

$C_m \cdot n\, C_{gs2} + C_{gs4} = 2C_{gs}$

HF Resp. of CM Loaded Differential Amp.

$$f_{p2} = \frac{g_{m3}}{2\pi C_{gs}} = \frac{g_{m3}}{2\pi \cdot 2C_{gs}} = \frac{f_T}{2}$$

$$f_{z2} \quad 2f_2 = f_T$$



CS, CG, SF, DA, CA

Wideband Amp. by Source/Emitter Degeneration

Wideband Amp. by Buffering the Input Signal

Wideband Amp. by CG/CB with an Input Buffer

Wideband Amp. by CG/CB with an Input Buffer

Template

Template